

SYSTEMS & PROGRAMMING

Python

Java / Apache Beam

SQL

Distributed Systems

API Services

Stream / Batch

Performance Testing

INFERENCE & LLM SYSTEMS

vLLM

PagedAttention

KV-cache

Continuous Batching

OpenAI-Compatible APIs

LoRA / QLoRA

PEFT

Transformers

4-bit / 8-bit Quantization

CLOUD & ORCHESTRATION

GCP

GKE

Vertex AI

Kubernetes

Docker

Helm

Istio

Cloud Build

ArgoCD

RELIABILITY & DELIVERY

CI/CD

GitOps

HPA Autoscaling

Prometheus

Grafana

Blue-Green Delivery

Load Testing

SLA Operations

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

Google Cloud
Certified Generative AI Leader

EDUCATION

B.E. Electrical & Electronics
Andhra University | 2012 - 2016

ML METHODS

Scikit-learn

PyTorch

TensorFlow

Keras

DenseNet

Classification

Predictive Modelling

Unsupervised Learning

Feature Engineering

Transfer Learning

Anomaly Detection

NLP

Time-Series Forecasting

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

EXPERIENCE

Lead Artificial Intelligence Engineer

Best Buy India

Jul 2025 - Present | Bengaluru

- > Architected a shared production platform for config-driven LoRA/QLoRA training, automated post-training evaluation, artifact lifecycle, and vLLM OpenAI-compatible model serving on GKE.
- > Enabled fine-tuned adapters to hot-load into a shared vLLM base model rather than require a separate deployment per model; self-hosted services support estimated annual API-cost savings of **~\$300K**. Kept model inputs company-managed and reduced external API exposure during peak demand.
- > Tuned vLLM batching, KV-cache and GPU-memory settings, 4-bit / 8-bit quantization, and parallelism from load-test evidence; owned HPA autoscaling, Istio traffic routing, blue-green releases, Prometheus, and Grafana.
- > Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.

Lead ML Engineer — Machine Learning & MLOps

Wipro

Aug 2023 - Jul 2025 | Hyderabad

- > Built distributed training and inference pipelines on NVIDIA DGX (8xA100), processing **billion-row** AWS CloudTrail datasets with Morpheus, RAPIDS, MLflow, continuous retraining, and Grafana alerting.
- > Engineered phishing classification at **2,400 emails/sec** - 2x prior throughput - and multiformat abuse-inbox processing integrated with Microsoft Sentinel and production security workflows.
- > Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; completed **300+ résumé screens** and conducted **50+ technical interviews**.

Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 - Sep 2023 | India

- > Led end-to-end Azure ML product delivery for BamaGruppen, increasing inspections from **130 to 160 trucks/day**, improving low-quality detection 17%, and generating **\$2.5M annual savings**.
- > Built production services and automation with Python, Azure Functions, Logic Apps, Azure DevOps, CI/CD, and Google OR-Tools; translated ambiguous client needs into operated systems.

PROJECTS

Shared LLM Training & Serving

Best Buy | GKE | vLLM | LoRA

A shared base-model approach for fine-tuning adapters, evaluation, artifact management, autoscaled inference, routing, safe rollout, and full-stack observability.

Real-Time Model Serving

Best Buy | Java / Beam | 40k feature events/sec

Streaming feature engineering and low-latency inference with Kubernetes autoscaling, secure endpoints, load testing, telemetry, GitOps delivery, and a 200ms SLA.

GPU STACK

- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Multi-GPU Training

OPTIMISATION & SIMULATION

- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Constraint Modelling
- Operations Research
- Assignment Models
- Supply-Chain Optimization
- Scenario Analysis

CLOUD & MANAGED ML

- GCS
- Artifact Registry
- Azure ML
- Azure DevOps
- Function Apps
- Logic Apps
- Google KCC

DATA & ML LIFECYCLE

- Apache Beam
- Dataflow
- Pub/Sub
- Vertex AI Feature Store
- Streaming Pipelines
- Training Pipelines
- Automated Evaluation
- Experiment Tracking
- Model Registry
- Model Versioning
- Continuous Training

LANGUAGES & DATA

- Pandas
- NumPy
- MS SQL Server
- IBM DB2
- Power BI

ENGINEERING LEADERSHIP

- Architecture Reviews
- Technical Mentoring
- Technical Hiring
- Team Building
- Stakeholder Alignment
- Cross-Functional Delivery

ACHIEVEMENTS

15 honors and awards across Best Buy, Wipro, Accenture, and Infosys.
Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.
Best Buy client AML Hall of Fame (2024): DGX delivery during the Wipro engagement.

ADDITIONAL PROJECTS & EXPERIENCE

GPU ML Pipelines

Wipro | DGX | Billion-row scale

Continuously retrained behavioral models and high-throughput classification pipelines with GPU acceleration, lifecycle tracking, monitoring, and analyst-facing integrations.

Inference Internals Learning Project

Personal project | PyTorch | In progress

Implementing KV-cache, paged-attention, and speculative-decoding concepts from scratch to deepen serving-system understanding; not presented as production framework work.

EARLIER EXPERIENCE & ACHIEVEMENTS

Data Scientist

GD Research Centre | May 2020 - May 2021

Built domain-aware time-series forecasting and scenario analysis across energy, manufacturing, healthcare, finance, retail, telecommunications, automotive, and other industry intelligence centers.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

Built an ML-powered lead-conversion prediction system and business-intelligence applications that moved sales prioritisation from intuition to data.

Achievements

15 honors and awards

Best Buy Talent Quotient (2026); Best Buy client AML Hall of Fame (2024), during the Wipro engagement.

Vehicle Routing & Transport Planning

Accenture | Google OR-Tools | Operations research

Designed vehicle-routing and mixed-integer models covering capacity, time windows, waiting time, demand, and departure constraints, then automated daily operational planning.

Industry Forecasting Platform

GD Research Centre | Time series | Scenario analysis

Designed and deployed domain-aware forecasting across 15+ industries, combining feature engineering, time-series modelling, and scenario analysis for enterprise decisions.

System Ownership

Design | Implementation | Production operations | Maintenance

Owned AI/ML systems from data and training pipelines through evaluation, deployment, monitoring, maintenance, incident support, and continuous improvement.

Engineering Leadership & Team Building

Architecture | Mentoring | Hiring | Cross-functional delivery

Stayed hands-on while leading ML/MLOps delivery, mentoring engineers, coordinating client and platform stakeholders, and helping scale the India AI engineering hub during the Wipro engagement through **300+ résumé screens** and **50+ technical interviews**.